

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
GitHub Repository	https://github.com/samyuktha-coder/ml/tree/main
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1 Objective

To implement and analyze the performance of three clustering algorithms – K-Means, DBSCAN (Density-Based Spatial Clustering of Applications with Noise), and Hierarchical Agglomerative Clustering (HAC) – on the Human Activity Recognition (HAR) Using Smartphones dataset. The clusters formed by each algorithm are visualized in two dimensions using PCA and t-SNE, compared against the ground-truth activity labels (although clustering itself is unsupervised), and evaluated using internal metrics (Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index) and external metrics (Adjusted Rand Index and Normalized Mutual Information).

2 Problem Statement

Problem: Group smartphone sensor windows into meaningful clusters without using activity labels during training, and determine which clustering algorithm (K-Means, DBSCAN, or Hierarchical) best recovers the structure of the six underlying human activities.

Input: The UCI Human Activity Recognition Using Smartphones dataset (training partition), containing 7352 windows of 2.56 s each (128 readings at 50 Hz) recorded from waist-mounted smartphone accelerometers and gyroscopes, each described by 561 time- and frequency-domain features.

Output: The K-Means elbow table (WCSS and Silhouette Score for $k = 2$ to 8), the selected value of k , tuned DBSCAN parameters (ϵ , minPts), a Ward-linkage hierarchical clustering with a comparison of four linkage criteria, 2D PCA/t-SNE cluster visualizations, dendrograms, a table and bar plots of internal and external evaluation metrics for all three algorithms, and a comparative analysis answering the observation questions.

3 DatasetDescription

Dataset Name	Human Activity Recognition Using Smartphones Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	7352 windows (training partition used)
Number of Features	561 (time- and frequency-domain features per window); reduced to 2 (PCA, for plotting) and 50 (PCA, for DBSCAN)

NumberOfClasses(groundtruth)	6 – WALKING, WALKINGUPSTAIRS, WALKINGDOWNSTAIRS, STANDING, SLEEPING, 3-axisaccelerometerand3-axisgyroscope
Sensors	sampled at 50 Hz 30volunteersaged19–48
Volunteers	2.56s(128readingsperwindow)
WindowSize	Colouringplotsandexternalevaluationonly
Labelsusedfor	(ARI, NMI) – never used to fit the clustering models

4 Exploratory Data Analysis

- Dataset overview (shape, label encoding)
- Feature distributions
- Ground-truth activities visualized using PCA and t-SNE

Figure: Feature Distributions

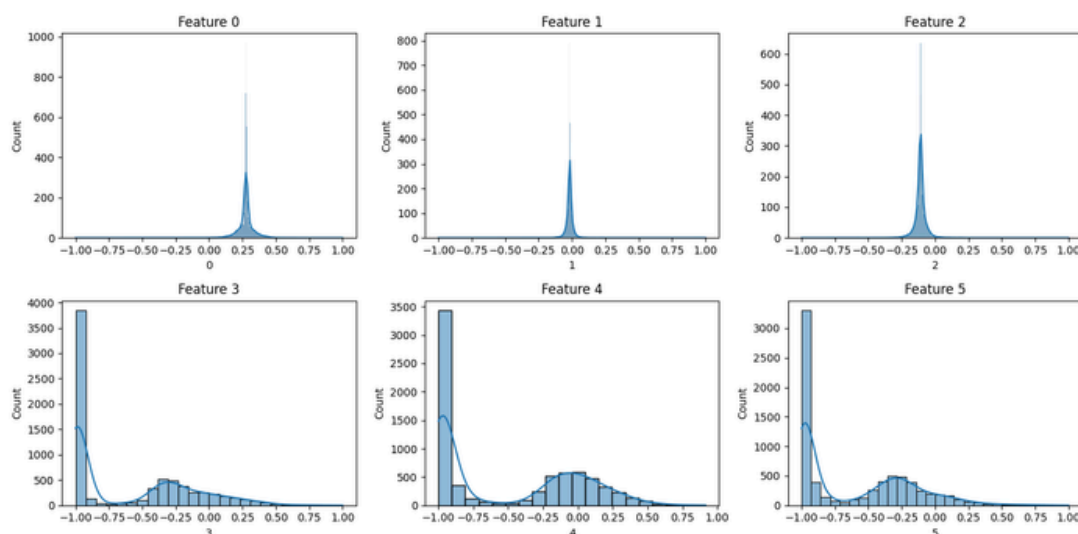


Figure 1: Histograms with KDE for the first six features of the raw dataset (features 0–5)

Inference: The first three features (mean body acceleration along X, Y and Z) are sharply peaked around a single value with a very narrow spread, so they are nearly constant across windows and carry little discriminative information on their own. The next three features (standard deviation of body acceleration along X, Y and Z) are clearly bimodal: a tall spike near -1 and a broader hump between roughly -0.5 and 0.5 . The spike near -1 corresponds to windows with almost no movement (sitting, standing, laying) while the broader hump corresponds to windows with body motion (walking activities). This bimodality is the first hint that the data has a natural two-group structure of static versus dynamic activities.

Figure: Ground-Truth Activities (PCA and t-SNE)

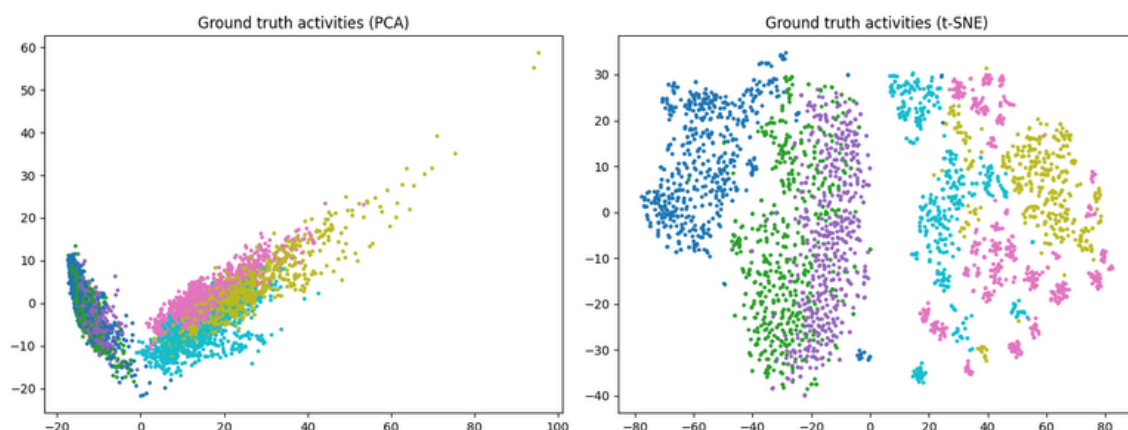


Figure 2: Ground-truth activity labels projected to 2D using PCA (all 7352 windows) and t-SNE (random subsample of 3000 windows)

Inference: In the t-SNE projection the data separates into two large, clearly distinct groups: the static activities (LAYING, SITTING, STANDING) on the left and the dynamic activities (WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS) on the right. Within the static group, SITTING and STANDING overlap heavily, while LAYING forms a somewhat separate region. Within the dynamic group, the three walking activities also overlap considerably. In the PCA projection the same picture appears in a more compressed form, with the static activities packed tightly on the left arm of a V-shaped cloud and the dynamic activities spread along the right arm. This shows that the dominant structure in the feature space is static versus dynamic, while the finer six-way activity distinction is much weaker – which directly affects how well any unsupervised algorithm can be expected to recover the six labels.

5 DataPreprocessing

- Missing value handling: Any missing entries are filled with the column median using `X.fillna(X.median())`, applied as a safeguard before scaling.
- Encoding: The activity labels (1–6) were mapped to their names using `activityLabels.txt` and then encoded with `LabelEncoder` into 0 = LAYING, 1 = SITTING, 2 = STANDING, 3 = WALKING, 4 = WALKING DOWNSTAIRS, 5 = WALKING UPSTAIRS. The encoded labels are used only for visualization and for computing ARI and NMI.
- Feature scaling: `StandardScaler` was applied to all 561 features (zero mean, unit variance), since K-Means, DBSCAN and Ward linkage are all based on Euclidean distance and would otherwise be dominated by large-magnitude features.
- Dimensionality reduction for visualization: PCA with 2 components (X2) for scatter plots, and t-SNE (perplexity=30, init="pca", random state=0) on a fixed random subsample of 3000 windows because t-SNE is computationally expensive.

- Dimensionality reduction for DBSCAN: PCA with 50 components (X50) was used as the input space for DBSCAN, because density estimation is unreliable in the full 561-dimensional space.
- Feature engineering: None – the 561 provided features were used directly.

6 Clustering Algorithms

K-Means Clustering

Working principle: Partition the data into k clusters by minimizing the within-cluster sum of squares, $WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$, using Euclidean distance. The algorithm initializes k centroids, assigns each point to its nearest centroid, recomputes each centroid as the mean of its assigned points, and repeats until convergence. Advantages: Simple, fast, and scales well to large datasets. Limitations: The number of clusters k must be specified in advance; assumes roughly spherical clusters of similar size; sensitive to initialization and outliers.

DBSCAN

Working principle: Density-based algorithm controlled by two parameters, the neighborhood radius ϵ and the minimum number of points minPts . Points with at least minPts neighbors within ϵ are core points; points within ϵ of a core point but not core themselves are border points; all remaining points are labelled noise (-1). Advantages: Does not require the number of clusters, can find clusters of arbitrary shape, and explicitly detects noise. Limitations: Very sensitive to ϵ and minPts ; struggles with clusters of varying density and with high-dimensional data.

Hierarchical Agglomerative Clustering (HAC)

Working principle: Bottom-up algorithm that starts with each point as its own cluster and repeatedly merges the two closest clusters according to a linkage criterion, producing a tree (dendrogram). Linkage options are single (minimum pairwise distance), complete (maximum pairwise distance), average (mean pairwise distance), and Ward's method (merge that gives the smallest increase in within-cluster variance). Advantages: No need to fix the number of clusters in advance; the dendrogram shows structure at all scales. Limitations: High memory and time cost on large datasets; merges cannot be undone; results depend strongly on the linkage choice.

7 Experimental Setup

Python Version	Python3 (Google Colab runtime)
Libraries	NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, SciPy
IDE	Google Colaboratory (Jupyter-based notebook)
Operating System	Linux (Google Colab backend)
Hardware	Google Colab-provided CPU runtime
Random Seed	0 (K-Means, t-SNE, subsampling)

8 K-Means with Elbow Method Results

K-Means was run for $k = 2$ to 8 with $n_{\text{init}}=10$ and $\text{random state}=0$ on the standardized 561-feature data. The WCSS (inertia) and Silhouette Score for each k are given in Table 2.

Table 2: K-Means Elbow Method Results		
Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	2336223.75	0.39
3	2081849.62	0.32
4	1976450.56	0.16
5	1875065.69	0.13
6	1818833.95	0.11
7	1776046.01	0.10
8	1735669.98	0.08

Figure: Elbow Curve and Silhouette Curve

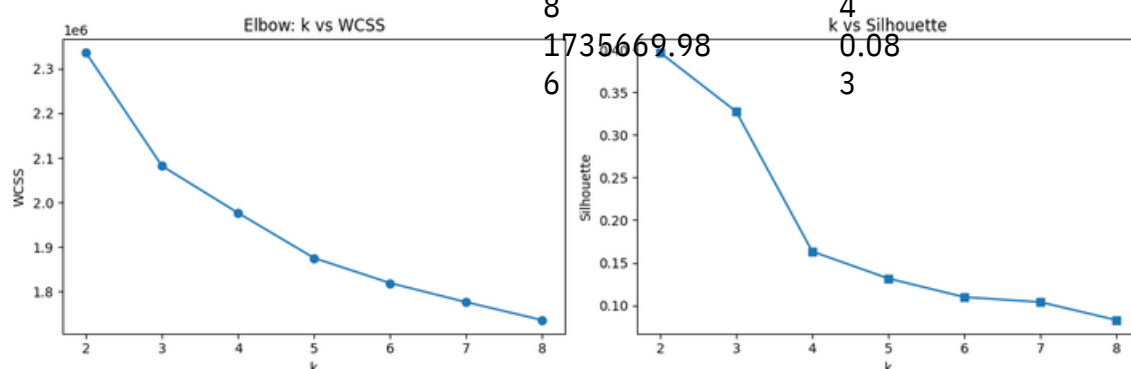
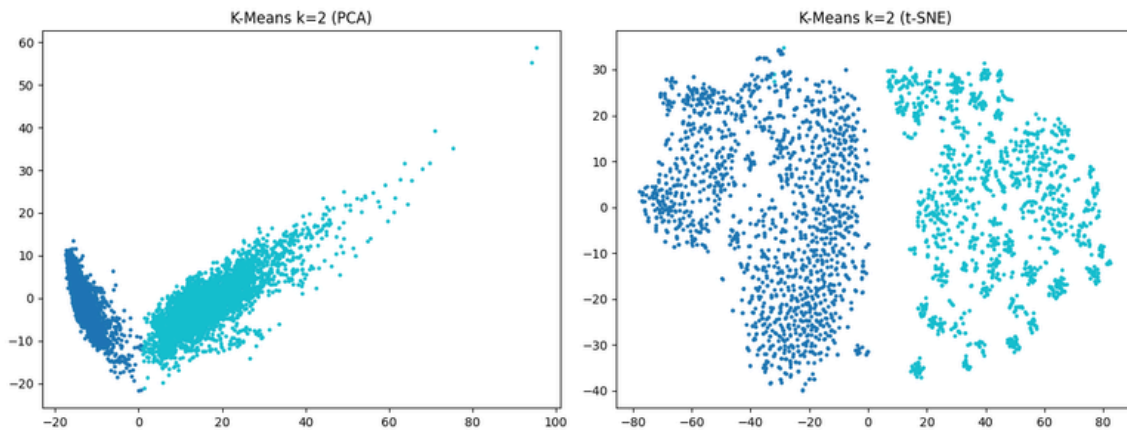


Figure 3: Left: elbow curve (k vs WCSS). Right: silhouette curve (k vs Silhouette Score)

Inference: WCSS decreases steadily as k grows, but the size of each drop shrinks: the largest reduction is from $k = 2$ to $k = 3$ (about 254,000), after which the drops fall to roughly 105,000, 101,000, 56,000, and then about 40,000–43,000 per step. The bend is gentle rather than sharp, with a mild elbow around $k = 3$. The silhouette curve is much clearer: it is highest at $k = 2$ (0.397), falls to 0.327 at $k = 3$, drops sharply to 0.163 at $k = 4$, and then keeps decreasing slowly to 0.083 at $k = 8$.

Selected k and justification: $k = 2$ was selected because it gives the highest silhouette score by a clear margin, and because the t-SNE projection of the ground truth shows two well-separated super-groups (static and dynamic activities). The elbow suggests $k = 3$ as a secondary candidate, but the silhouette score is already noticeably lower there. Choosing $k = 6$ to match the six activity labels would have given a silhouette of only 0.110, showing that six compact, well-separated clusters do not exist in this feature space.

Figure: K-Means Clusters ($k = 2$)Figure 4: K-Means clusters for $k = 2$ visualized with PCA (left) and t-SNE (right)

Inference: K-Means with $k = 2$ splits the data almost exactly along the static/dynamic boundary seen in the ground-truth plot: one cluster covers the compact static region on the left of the PCA plot and the left group in t-SNE, and the other covers the dynamic activities on the right. The two clusters are clean and well separated in t-SNE, but the model cannot distinguish sitting from standing from laying, or the three walking types from each other, since these are merged into the two clusters.

9 Clustering Implementation

9.1 K-Means

K-Means was fitted for $k = 2, \dots, 8$ on the standardized data (StandardScaler, 561 features), and the model with the highest silhouette score ($k = 2$) was retained as the final K-Means model. Results are reported in Section 8.

9.2 DBSCAN

DBSCAN was run on the 50-component PCA representation. To choose ϵ , the distance of every point to its 10th nearest neighbour was computed and sorted (k-distance graph).

Candidate values of ϵ were taken as the 70th, 80th, 90th and 95th percentiles of these distances, and each was combined with $\text{minPts} \in \{5, 10, 20\}$, giving 12 configurations. Configurations with at least two clusters and less than 50% noise were compared using the silhouette score computed on non-noise points, and the best was selected.

Figure: k-Distance Graph

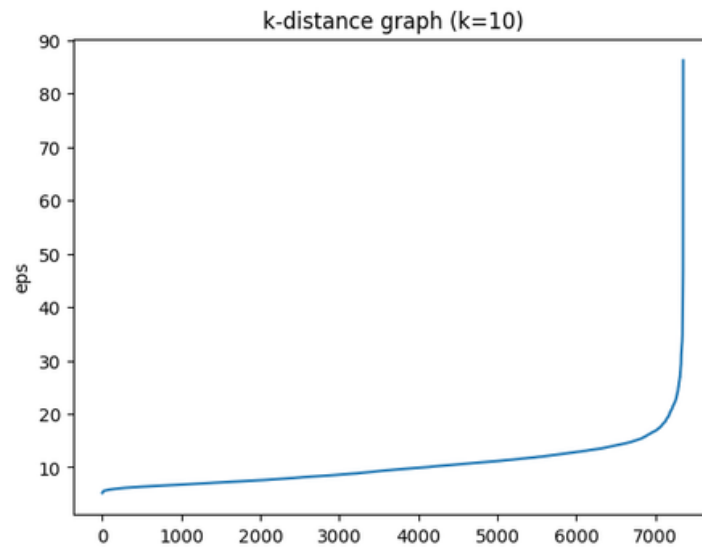


Figure 5: Sorted distance to the 10th nearest neighbour for every point (k-distance graph)

Inference: The k-distance curve rises slowly from about 5 to about 15 across roughly 90% of the points and then shoots upward sharply for the last few hundred points, reaching values above 80. The knee therefore lies at a distance of roughly 15–20, and the points above it are isolated windows that DBSCAN will treat as noise for any reasonable ϵ . The candidate ϵ values (11.35, 12.58, 14.45, 16.70) cover the region up to and around this knee.

Table 3: DBSCAN Parameter Tuning (PCA-50 space)

ϵ	minPts	Clusters	NoisePoints	Noise(%)	Silhouette(non-noise)
11.346	5	18	1154	15.7	0.349
11.346	10	9	1411	19.2	0.385
11.346	20	3	1762	24.0	0.373
12.581	5	7	653	8.9	0.051
12.581	10	6	830	11.3	0.225
12.581	20	4	1140	15.5	0.306
14.448	5	2	298	4.1	0.381
14.448	10	2	385	5.2	0.339
14.448	20	2	498	6.8	0.339
16.695	5	1	171	2.3	–
16.695	10	1	191	2.6	–
16.695	20	1	230	3.1	–

Selected parameters: $\epsilon = 11.35$, minPts = 10, giving 9 clusters and 1411 noise points (19.2% of the data) with a non-noise silhouette of 0.385.

Figure: DBSCAN Clusters

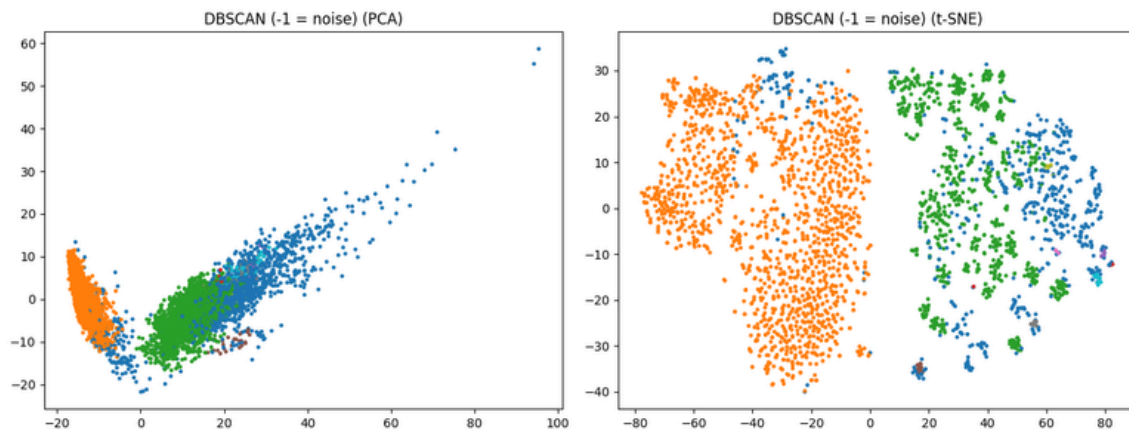


Figure 6: DBSCAN clusters ($\epsilon = 11.35$, $\text{minPts} = 10$) visualized with PCA (left) and t-SNE (right). Label -1 (noise) is the lowest colour value and therefore appears in blue

Inference: DBSCAN also recovers the static/dynamic split: one large cluster covers the dense static region (orange), and a second large cluster covers most of the dense part of the dynamic region (green). The other clusters are very small and appear only as small patches. The noise points are mostly located in the sparse parts of the dynamic region, which makes sense because windows with unusual or irregular motion are less dense than those of stationary activities. The tuning table shows that DBSCAN is very sensitive to its parameters: at $\epsilon = 11.35$ increasing minPts from 5 to 20 changes the cluster count from 18 to 3 and the noise from 1154 to 1762 points, while at $\epsilon = 16.70$ everything merges into a single cluster.

9.3 HierarchicalClustering

Agglomerative clustering was applied on the standardized 561-feature data with the number of clusters fixed to 6 (the number of true activities). Ward's linkage was used as the main model, as required, and single, complete and average linkage were also run to study the effect of linkage choice.

Table 4: Effect of Linkage Criterion on Hierarchical Clustering ($n_{\text{clusters}} = 6$)

Linkage	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
Ward	0.083	2.786	1730.836	0.279	0.454
Complete	0.304	1.401	506.998	0.048	0.148
Average	0.516	0.655	32.593	0.000	0.003
Single	0.545	0.281	26.111	0.000	0.002

Figure: HAC (Ward) Clusters

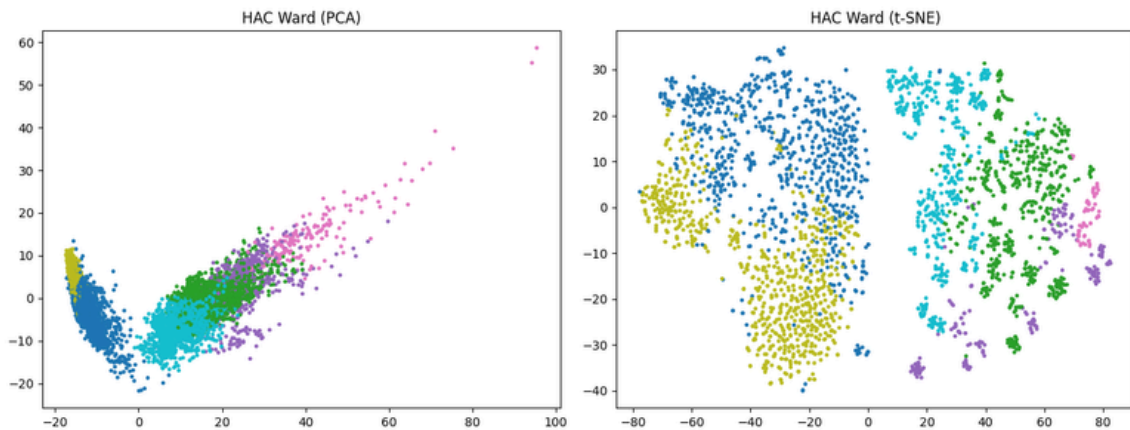


Figure 7: Hierarchical clustering with Ward linkage (n clusters = 6) visualized with PCA (left) and t-SNE (right)

Inference: Ward's method splits the data into six clusters. In the t-SNE plot two clusters fall in the static group on the left and four in the dynamic group on the right, so unlike K-Means it attempts to separate finer structure inside each super-group. The boundaries between neighbouring clusters overlap, in line with the overlap of the true activities seen in the ground-truth plot, which explains the low silhouette score of 0.083 for this model.

Figure: Dendrograms for Different Linkages

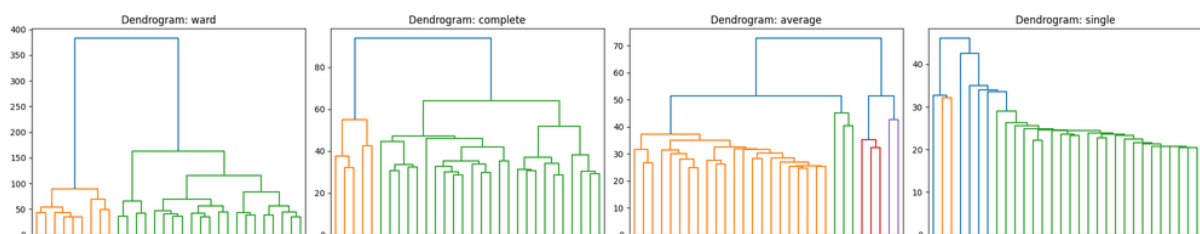


Figure 8: Truncated dendrograms (last 30 merges) on a 300-window subsample for Ward, complete, average and single linkage

Inference: The Ward dendrogram has a very tall final merge (height around 380) joining two main branches, followed by a second large merge (height around 165), which points to two dominant groups (static and dynamic) with further sub-structure below. The complete-linkage dendrogram is more compact (final merge around 94) with a smaller isolated branch. The average-linkage dendrogram has one very large branch plus a few small, late-merging branches, and the single-linkage dendrogram shows the typical chain-linking pattern: a staircase of merges at almost equal, low heights, where points are absorbed one by one into a growing cluster.

10 Evaluation Metrics and Results

Internal metrics (no labels): Silhouette Score (higher is better), Davies–Bouldin Index (lower is better), Calinski–Harabasz Index (higher is better). External metrics (against true activities): Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI), both higher is better. For DBSCAN, internal metrics are computed on non-noise points in the PCA-50 space it was fitted on, whereas ARI and NMI use all points with noise treated as its own label; for K-Means and HAC the internal metrics are computed in the full standardized space.

Table 5: Comparison of the Three Clustering Algorithms

Model	Clusters	Noise	Silhouette	Davies–Bouldin	Calinski–Harabasz	ARI	NMI
K-Means (k = 2)	2	0	0.397	1.069	5626.013	0.328	0.543
DBSCAN	9	1411	0.385	1.115	962.946	0.324	0.479
HAC (Ward, 6)	6	0	0.083	2.786	1730.836	0.279	0.454

Best model overall: K-Means with k = 2 (best on all five metrics).

Figure: Metric Comparison

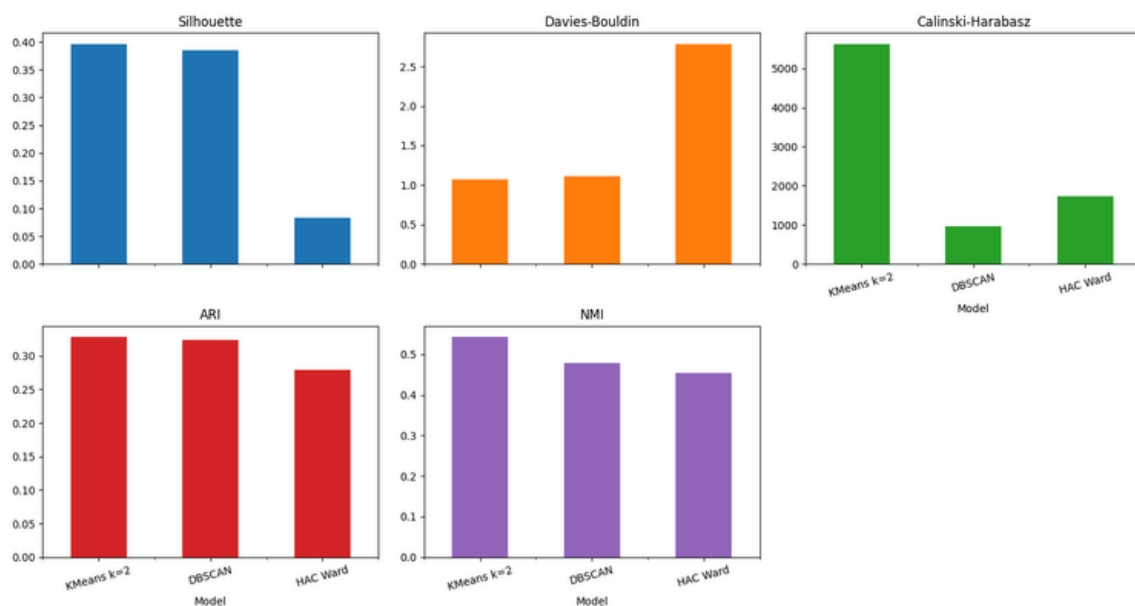


Figure 9: Bar plots of Silhouette, Davies–Bouldin, Calinski–Harabasz, ARI and NMI for K-Means, DBSCAN and HAC (Ward)

Inference: K-Means (k = 2) has the highest silhouette (0.397), the lowest Davies–Bouldin (1.069), the highest Calinski–Harabasz (5626) and the highest ARI (0.328) and NMI (0.543). DBSCAN is very close to K-Means on silhouette, Davies–Bouldin and ARI (0.385, 1.115, 0.324), which is expected since it found essentially the same static/dynamic split, but it lags on NMI (0.479) and on Calinski–Harabasz (963). HAC with Ward linkage is clearly weakest on the internal metrics (silhouette 0.083, Davies–Bouldin 2.786), although its ARI (0.279) and NMI (0.454) are not far behind, because it is the only model

that tries to split the data into six groups. Note that the absolute ARI values (0.28–0.33) are modest for all three algorithms, confirming that none of them recovers the six activities cleanly.

11 Observations and Comparative Analysis

Which algorithm produced the most meaningful clusters? Why? K-Means with

$k = 2$ produced the most meaningful clusters: it scored best on all five metrics and gave a clean, well-separated split of static versus dynamic activities that matches the main structure visible in the t-SNE ground-truth plot. DBSCAN was a close second and found the same two dense regions, while additionally marking 19.2% of the windows as noise. Ward-linkage HAC was the weakest on internal measures. It should be noted, however, that “most meaningful” here means recovering the dominant two-group structure; none of the algorithms separated the six individual activities well (ARI at most 0.328). The silhouette score dropped from 0.397 at $k = 2$ to 0.327 at $k = 3$ and to 0.163 at $k = 4$, and continued down to 0.083 at $k = 8$, so quality falls quickly once k exceeds the natural two-group structure. The WCSS curve, by contrast, decreases smoothly with no sharp elbow (only a mild bend near $k = 3$), so the elbow method alone would not give a decisive choice on this dataset and needs to be combined with the silhouette curve.

Did DBSCAN detect noise or small clusters effectively? Partly. With $\epsilon = 11.35$ and $\text{minPts} = 10$ it flagged 1411 points (19.2%) as noise, mainly sparse, irregular windows in the dynamic region, and produced several very small clusters in addition to the two large ones. However, the results depended strongly on parameters: the number of clusters ranged from 18 to 1 and the noise from 171 to 1762 points across the 12 tested settings, and the small clusters do not correspond to individual activities. DBSCAN is therefore useful for identifying outlier windows but not for separating the six activities.

How does linkage choice (single/complete/Ward) affect hierarchical clustering? Very strongly. Ward linkage gave the best agreement with the true labels (ARI 0.279, NMI 0.454), complete linkage gave weaker agreement (ARI 0.048, NMI 0.148), and average and single linkage gave essentially no agreement (ARI 0.000, NMI 0.003 and 0.002). The single-linkage dendrogram shows the chaining effect (points absorbed one by one into a growing cluster), and the ARI and NMI values close to zero for single and average linkage indicate that the six-cluster cuts of these trees do not divide the data in a way related to the activities, most likely isolating a few outliers while leaving nearly all points in one cluster.

Which internal metric best matched your visual intuition of cluster quality?

The Calinski–Harabasz index. Across the four linkage criteria it ranked the models in exactly the same order as the external metrics and as visual inspection (Ward 1730.8 > complete 507.0 > average 32.6 > single 26.1). Silhouette and Davies–Bouldin were misleading in that comparison: they gave their best values to single (0.545, 0.281) and average (0.516, 0.655) linkage even though those partitions have essentially zero agreement with the activity labels and are visually meaningless. Silhouette agreed well with intuition for the K-Means k selection, where its peak at $k = 2$ matched the two visible groups.

12 Conclusion

This experiment applied K-Means, DBSCAN and Hierarchical Agglomerative Clustering to the UCI Human Activity Recognition dataset (7352 windows, 561 standardized features). K-Means with $k = 2$, chosen by the silhouette peak (0.397) and supported by the mild WCSS elbow near $k = 3$, gave the best results on all metrics (silhouette 0.397, Davies–Bouldin 1.069, Calinski–Harabasz 5626, ARI 0.328, NMI 0.543). DBSCAN ($\epsilon = 11.35$, $\text{minPts} = 10$) reached similar scores (silhouette 0.385, ARI 0.324) and additionally identified 1411 noise points, but was highly sensitive to its parameters.

Ward-linkage HAC with six clusters attempted a finer split and reached ARI 0.279 and NMI 0.454, while single and average linkage failed completely (ARI ≈ 0). The PCA and t-SNE plots show that the dominant structure in the data is static versus dynamic activity, while sitting/standing and the three walking activities overlap heavily, so all three unsupervised methods recover the two-group structure far better than the six individual activities. The linkage comparison also showed that Calinski–Harabasz tracked the external metrics most reliably, whereas silhouette and Davies–Bouldin can be misleading for degenerate partitions.

13 References

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